

The Open University of Sri Lanka
Faculty of Engineering Technology
Bachelor of Software Engineering Honours
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EEI4366 – Mini Project

Hospital Management System Database

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<Q1 – 523607238>

```
1  -- Mini Project SQL Script
2  -- Registration Number: 523607238
3
4 • use HospitalDB;
5
6  -- Q1 - Create patient surgery view
7
8  -- Create view to display patient and surgery information
9 • create view PatientSurgeryView as
10 select
11     p.Patient_ID as 'Patient_ID',
12     p.Name as 'Patient_Name',
13     concat(l.Bed_No, ' - ', l.Room_No) as 'Location',
14     s.Surgery_Name as 'Surgery_Name',
15     s.Date as 'Surgery_Date'
16 from Patient p
17 join Location l on p.Location_ID = l.Location_ID
18 join Surgery s on p.Patient_ID = s.Patient_ID;
19
```

Output				
Action Output				
#	Time	Action	Message	
✓ 1	14:35:41	use HospitalDB	0 row(s) affected	
✓ 2	14:35:41	create view PatientSurgeryView as select p.Patient_ID as 'Patient_ID', p.Name as 'Patient_Name', co...	0 row(s) affected	

Result:

```
20  -- Test the view
21 • select * from PatientSurgeryView;
```

Result Grid					
		Filter Rows:	Export:	Wrap Cell Content: fA	
	Patient_ID	Patient_Name	Location	Surgery_Name	Surgery_Date
▶	PT001	Sahan Perera	B12 - R101	Heart Bypass	2025-03-01
	PT002	Nadeesha Kumari	B09 - R102	Brain Tumor Removal	2025-03-03
	PT003	Milan Jayasuriya	B05 - R201	Bone Fixation	2025-03-05
	PT004	Ayesha Fernando	B14 - R103	Skin Graft	2025-03-07
	PT005	Thilina Dissanayake	B07 - R202	Appendix Removal	2025-03-09

SurgeryView 1 ×

Output				
Action Output				
#	Time	Action	Message	
✓ 2	14:35:41	create view PatientSurgeryView as select p.Patient_ID as 'Patient_ID', p.Name as 'Patient_Name', co...	0 row(s) affected	
✓ 3	14:38:05	select * from PatientSurgeryView LIMIT 0, 1000	5 row(s) returned	

<Q2 – 523607238>

```
25  -- create MedInfo table
26 • create table MedInfo (
27      MedName varchar(100) primary key,
28      QuantityAvailable int check (QuantityAvailable >= 0),
29      ExpirationDate date
30 );
31
```

Output

#	Time	Action	Message
3	14:50:50	select *from PatientSurgeryView LIMIT 0, 1000	5 row(s) returned
4	14:51:35	create table MedInfo (MedName varchar(100) primary key, QuantityAvailable int check (QuantityAvailabl...	0 row(s) affected

i.

```
32  -- i. Create trigger for insert operation
33  delimiter $$
34 • create trigger after_medication_insert
35  after insert on Medication
36  for each row
37  begin
38      insert into MedInfo (MedName, QuantityAvailable, ExpirationDate)
39      values (new.Name, new.Quantity_On_Hand, new.Expiration_Date);
40  end$$
41  delimiter ;
42
```

Output

#	Time	Action	Message
4	14:51:35	create table MedInfo (MedName varchar(100) primary key, QuantityAvailable int check (QuantityAvailabl...	0 row(s) affected
5	14:53:48	create trigger after_medication_insert after insert on Medication for each row begin insert into MedInfo (Me...	0 row(s) affected

Result:

```
32  -- i. Create trigger for insert operation
33  delimiter $$
34 • create trigger after_medication_insert
35  after insert on Medication
36  for each row
37  begin
38      insert into MedInfo (MedName, QuantityAvailable, ExpirationDate)
39      values (new.Name, new.Quantity_On_Hand, new.Expiration_Date);
40  end$$
41  delimiter ;
42
```

Output

#	Time	Action	Message
4	14:51:35	create table MedInfo (MedName varchar(100) primary key, QuantityAvailable int check (QuantityAvailabl...	0 row(s) affected
5	14:53:48	create trigger after_medication_insert after insert on Medication for each row begin insert into MedInfo (Me...	0 row(s) affected

ii.

```
50  -- ii. Create trigger for update operation
51  delimiter $$
52 • create trigger after_medication_update
53  after update on Medication
54  for each row
55  begin
56      update MedInfo
57      set QuantityAvailable = new.Quantity_On_Hand,
58          ExpirationDate = new.Expiration_Date
59      where MedName = old.Name;
60  end$$
61  delimiter ;
62
```

Output

#	Time	Action	Message
7	14:55:36	select * from MedInfo LIMIT 0, 1000	1 row(s) returned
8	15:22:41	create trigger after_medication_update after update on Medication for each row begin update MedInfo ...	0 row(s) affected

Result:

```
63 • -- Test update trigger
64 update Medication
65 set Quantity_On_Hand = 300, Expiration_Date = '2026-12-31'
66 where Medication_Code = 'MD006';
67
68 -- Check MedInfo table after update
69 • select * from MedInfo where MedName = 'Ibuprofen';
70
```

MedName	QuantityAvailable	ExpirationDate
Ibuprofen	300	2026-12-31
NULL	NULL	NULL

MedInfo 3 x

Output

Action Output

#	Time	Action	Message
9	15:24:36	update Medication set Quantity_On_Hand = 300, Expiration_Date = '2026-12-31' where Medication_Code = '...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
10	15:24:36	select * from MedInfo where MedName = 'Ibuprofen' LIMIT 0, 1000	1 row(s) returned

iii.

```
71 -- iii. Create trigger for delete operation
72 delimiter $$
73 • create trigger after_medication_delete
74 after delete on Medication
75 for each row
76 begin
77     delete from MedInfo where MedName = old.Name;
78 end$$
79 delimiter ;
80
```

Output

Action Output

#	Time	Action	Message
10	15:24:36	select * from MedInfo where MedName = 'Ibuprofen' LIMIT 0, 1000	1 row(s) returned
11	15:25:39	create trigger after_medication_delete after delete on Medication for each row begin delete from MedInfo ...	0 row(s) affected

Result:

```
81 • -- Test delete trigger
82   delete from Medication where Medication_Code = 'MD006';
83
84   -- Check MedInfo table after delete
85 • select * from MedInfo where MedName = 'Ibuprofen';
86
```

Result Grid				Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:
	MedName	QuantityAvailable	ExpirationDate				
*	NULL	NULL	NULL				

MedInfo 4 x			
Output			
Action Output			
#	Time	Action	Message
✓ 12	15:26:33	delete from Medication where Medication_Code = 'MD006'	1 row(s) affected
✓ 13	15:26:33	select * from MedInfo where MedName = 'Ibuprofen' LIMIT 0, 1000	0 row(s) returned

<Q3-523607238>

```
89   -- Create stored procedure to count patient medications
90   delimiter $$
91 • create procedure GetPatientMedicationCount(
92     in patient_id_param varchar(10),
93     out med_count int
94   )
95   begin
96     select count(*) into med_count
97     from Patient_Medication
98     where Patient_ID = patient_id_param;
99   end$$
100  delimiter ;
101
```

Output			
Action Output			
#	Time	Action	Message
✓ 13	15:26:33	select * from MedInfo where MedName = 'Ibuprofen' LIMIT 0, 1000	0 row(s) returned
✓ 14	15:42:24	create procedure GetPatientMedicationCount(in patient_id_param varchar(10), out med_count int) begin...	0 row(s) affected

Result:

```
102 • -- Test the stored procedure
103   set @result = 0;
104 • call GetPatientMedicationCount('PT001', @result);
105 • select @result as 'Medication_Count_PT001';
106
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	Medication_Count_PT001
▶	1

Result 5 x

Output

Action Output

#	Time	Action	Message
✓ 16	15:44:44	call GetPatientMedicationCount('PT001', @result)	1 row(s) affected
✓ 17	15:44:44	select @result as 'Medication_Count_PT001' LIMIT 0, 1000	1 row(s) returned

```
107 -- Test with another patient
108 • call GetPatientMedicationCount('PT003', @result);
109 • select @result as 'Medication_Count_PT003';
110
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	Medication_Count_PT003
▶	1

Result 6 x

Output

Action Output

#	Time	Action	Message
✓ 18	15:46:06	call GetPatientMedicationCount('PT003', @result)	1 row(s) affected
✓ 19	15:46:06	select @result as 'Medication_Count_PT003' LIMIT 0, 1000	1 row(s) returned

<Q4 – 523607238>

```
113  -- Create function to calculate days to expiry
114  delimiter $$
115 • create function DaysToExpiry(exp_date date)
116  returns int
117  deterministic
118  begin
119      return datediff(exp_date, curdate());
120  end$$
121  delimiter ;
122
```

Output

Action Output

#	Time	Action	Message
✓ 19	15:46:06	select @result as 'Medication_Count_PT003' LIMIT 0, 1000	1 row(s) returned
✓ 20	15:48:15	create function DaysToExpiry(exp_date date) returns int deterministic begin return datediff(exp_date, curdat...	0 row(s) affected

```
123 • -- Create view for expiring medications
124  create view ExpiringMedications as
125  select
126      Medication_Code,
127      Name as Medication_Name,
128      Quantity_On_Hand,
129      Expiration_Date,
130      DaysToExpiry(Expiration_Date) as Days_Remaining
131  from Medication
132  where DaysToExpiry(Expiration_Date) < 30
133         and DaysToExpiry(Expiration_Date) > 0;
134
```

Output

Action Output

#	Time	Action	Message
✓ 20	15:48:15	create function DaysToExpiry(exp_date date) returns int deterministic begin return datediff(exp_date, curdat...	0 row(s) affected
✓ 21	15:51:40	create view ExpiringMedications as select Medication_Code, Name as Medication_Name, Quantity_O...	0 row(s) affected

Result:

```
135 -- Insert a test medication that expires soon
136 • insert into Medication values
137 ('MD007', 'Test Medicine', 100, date_add(curdate(), interval 15 day), 20.00);
138
139 -- Show expiring medications
140 • select * from ExpiringMedications;
```

1/1

Medication_Code	Medication_Name	Quantity_On_Hand	Expiration_Date	Days_Remaining
MD001	Amoxicillin	200	2026-01-10	26
MD007	Test Medicine	100	2025-12-30	15

ExpiringMedications7 x

Output

#	Time	Action	Message
✓ 22	15:53:10	insert into Medication values ('MD007', 'Test Medicine', 100, date_add(curdate(), interval 15 day), 20.00)	1 row(s) affected
✓ 23	15:53:11	select * from ExpiringMedications LIMIT 0, 1000	2 row(s) returned

<Q6 – 523607238>

i. **staff.xml file**

```
<Staff>
  <row>
    <Employee_No>ST010</Employee_No>
    <Name>Dr. Sunil Fernando</Name>
    <Gender>Male</Gender>
    <Address>Kandy</Address>
    <Telephone_No>0771111222</Telephone_No>
    <Role>Doctor</Role>
  </row>
  <row>
    <Employee_No>ST011</Employee_No>
    <Name>Nurse Anoma</Name>
    <Gender>Female</Gender>
    <Address>Galle</Address>
    <Telephone_No>0772222333</Telephone_No>
    <Role>Nurse</Role>
  </row>
  <row>
    <Employee_No>ST012</Employee_No>
    <Name>Surgeon Kamal</Name>
    <Gender>Male</Gender>
    <Address>Colombo</Address>
    <Telephone_No>0773333444</Telephone_No>
    <Role>Surgeon</Role>
  </row>
</Staff>
```

```

145 • load xml
146 infile 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/staff.xml'
147 into table Staff
148 rows identified by '<row>';
149

```

Output

#	Time	Action	Message
23	15:53:11	select * from ExpiringMedications LIMIT 0, 1000	2 row(s) returned
24	16:00:18	load xml infile 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/staff.xml' into table Staff rows identified b...	3 row(s) affected Records: 3 Deleted: 0 Skipped: 0 Warnings: 0

Result:

```

150 -- Check results
151 • select * from Staff;
--

```

Result Grid

Employee_No	Name	Gender	Address	Telephone_No	Role
ST001	Dr. Amal Perera	Male	Colombo	0771234567	Doctor
ST002	Dr. Nadeesha Silva	Female	Kandy	0779876543	Doctor
ST003	Dr. Kasun Jayasuriya	Male	Galle	0714567890	Doctor
ST004	Nurse Sanduni	Female	Kurunegala	0751122334	Nurse
ST005	Nurse Ruwan	Male	Negombo	0709988776	Nurse
ST006	Surgeon Tharindu	Male	Matara	0766655443	Surgeon
ST007	Surgeon Dilki	Female	Jaffna	0728899001	Surgeon
ST010	Dr. Sunil Fernando	Male	Kandy	0771111222	Doctor
ST011	Nurse Anoma	Female	Galle	0772222333	Nurse
ST012	Surgeon Kamal	Male	Colombo	0773333444	Surgeon

Staff 8

Output

#	Time	Action	Message
24	16:00:18	load xml infile 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/staff.xml' into table Staff rows identified b...	3 row(s) affected Records: 3 Deleted: 0 Skipped: 0 Warnings: 0
25	16:00:57	select * from Staff LIMIT 0, 1000	10 row(s) returned

ii. patient.xml file

```
<Patient>
  <row>
    <Patient_ID>PT008</Patient_ID>
    <Name>Rajitha Perera</Name>
    <Age>35</Age>
    <Address>Negombo</Address>
    <Telephone_No>0774444555</Telephone_No>
    <Blood_Type>O+</Blood_Type>
    <Allergy_ID>AL003</Allergy_ID>
    <Location_ID>L001</Location_ID>
  </row>
  <row>
    <Patient_ID>PT009</Patient_ID>
    <Name>Sarojini Silva</Name>
    <Age>42</Age>
    <Address>Matara</Address>
    <Telephone_No>0775555666</Telephone_No>
    <Blood_Type>A+</Blood_Type>
    <Allergy_ID>AL002</Allergy_ID>
    <Location_ID>L002</Location_ID>
  </row>
  <row>
    <Patient_ID>PT010</Patient_ID>
    <Name>Lakshitha Bandara</Name>
    <Age>29</Age>
    <Address>Kurunegala</Address>
    <Telephone_No>0776666777</Telephone_No>
    <Blood_Type>B+</Blood_Type>
    <Allergy_ID>AL001</Allergy_ID>
    <Location_ID>L003</Location_ID>
  </row>
</Patient>
```

```
153  -- Load Patient XML
154 • load xml
155  infile 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/patient.xml'
156  into table Patient
157  rows identified by '<row>';
158
```

Output			
Action Output			
#	Time	Action	Message
✓ 25	16:00:57	select * from Staff LIMIT 0, 1000	10 row(s) returned
✓ 26	16:12:16	load xml infile 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/patient.xml' into table Patient rows identifi...	3 row(s) affected Records: 3 Deleted: 0 Skipped: 0 Warnings: 0

Result:

```
159 -- Check results
160 • select * from Patient;
```

Patient_ID	Name	Age	Address	Telephone_No	Blood_Type	Allergy_ID	Location_ID
PT001	Sahan Perera	25	Colombo	0771112223	A+	AL001	L001
PT002	Nadeesha Kumari	30	Kandy	0772223334	B+	AL003	L002
PT003	Milan Jayasuriya	40	Galle	0713334445	O-	AL002	L003
PT004	Ayesha Fernando	28	Negombo	0704445556	AB+	AL005	L004
PT005	Thilina Dissanayake	22	Matara	0755556667	A-	AL004	L005
PT008	Rajitha Perera	35	Negombo	0774444555	O+	AL003	L001
PT009	Sarojini Silva	42	Matara	0775555666	A+	AL002	L002
PT010	Lakshitha Bandara	29	Kurunegala	0776666777	B+	AL001	L003
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Patient 9 x

Output

#	Time	Action	Message
✓ 26	16:12:16	load xml infile 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/patient.xml' into table Patient rows identi...	3 row(s) affected Records: 3 Deleted: 0 Skipped: 0 Warnings: 0
✓ 27	16:13:11	select * from Patient LIMIT 0, 1000	8 row(s) returned
